MLCHIP 2025-Spring Final Project
AXI Introduction

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Outline

- Bus Architecture
- Handshake Process
- AXI Introduction
- Read Transaction
- Write Transaction
- Reference

Bus Architecture

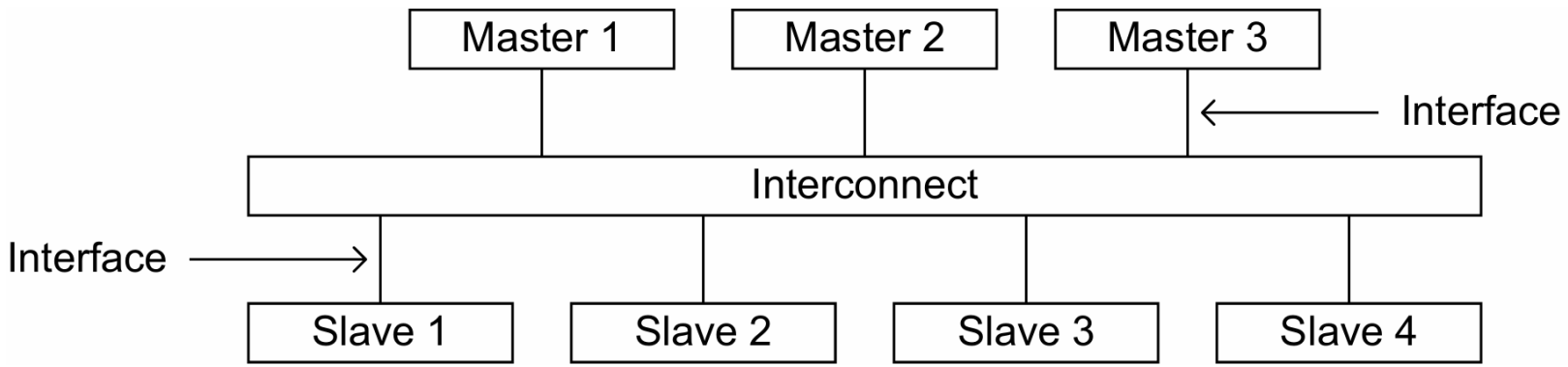
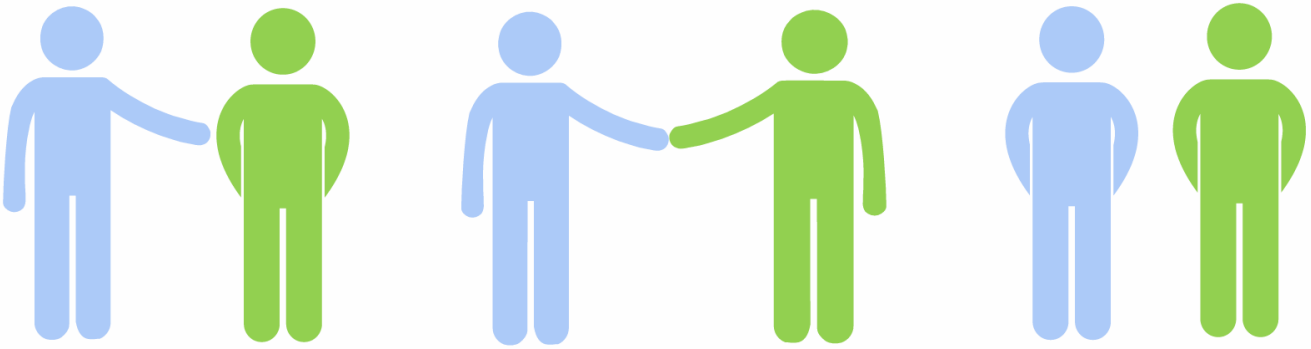


Figure A1-3 Interface and interconnect

Handshake Process Scenario 1

- Slave: Valid Master: Ready



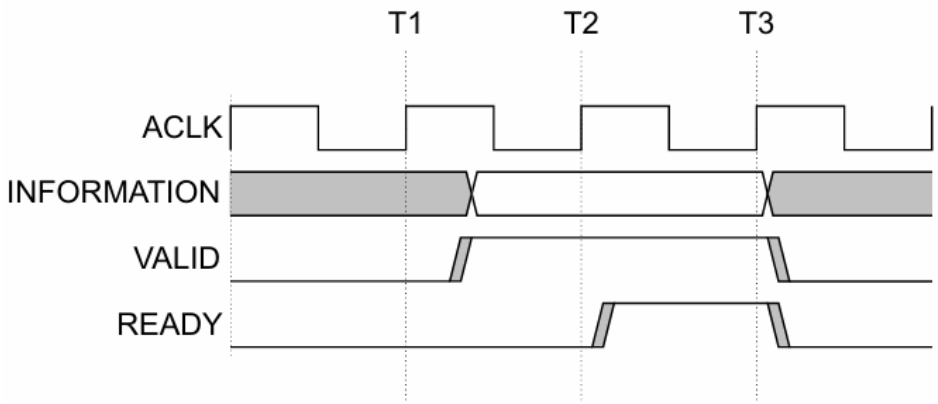
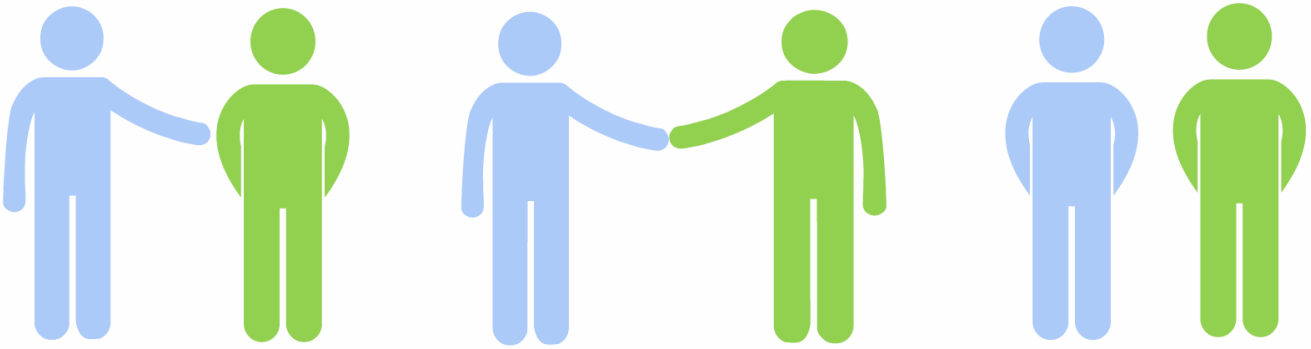


Figure A3-2 VALID before READY handshake

Handshake Process Scenario 2

- Slave: Valid Master: Ready



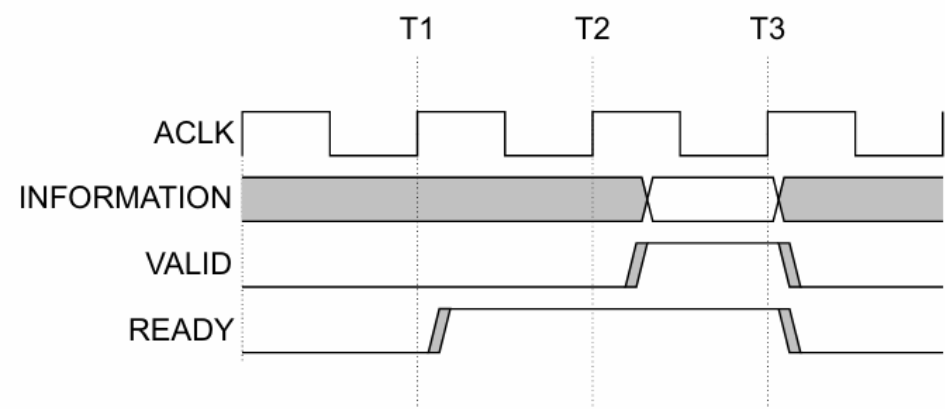
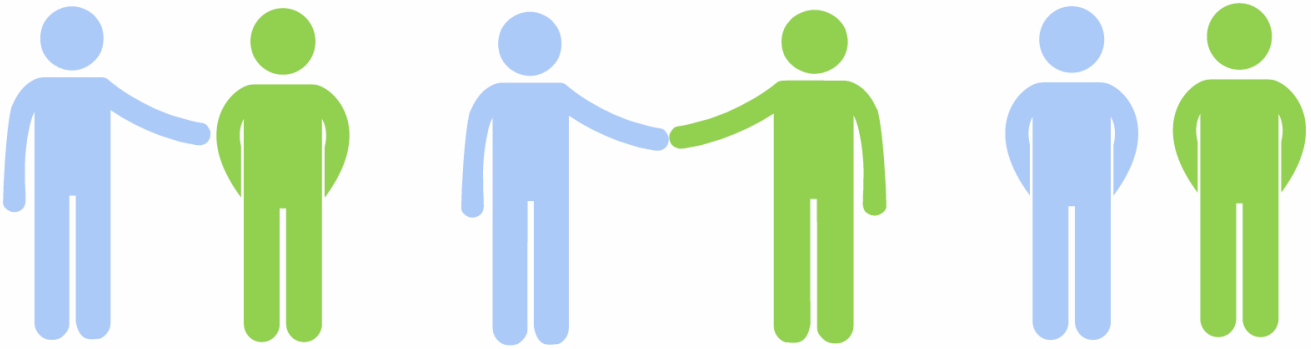


Figure A3-3 READY before VALID handshake

Handshake Process Scenario 3

- Slave: Valid Master: Ready



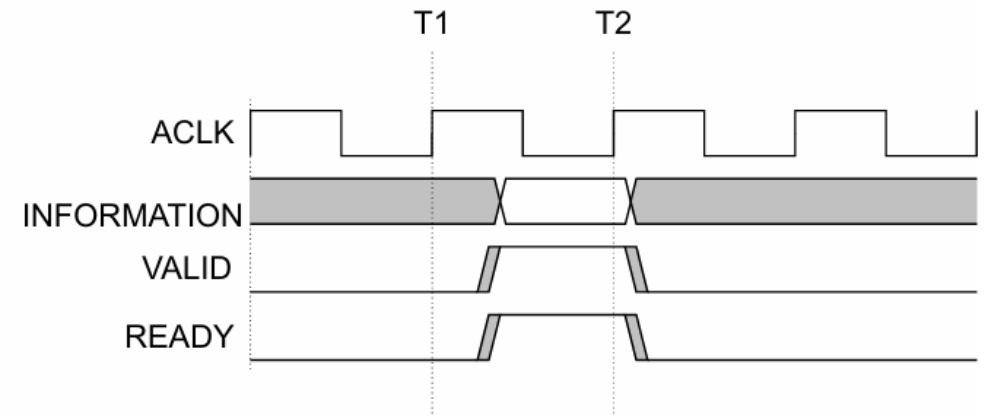


Figure A3-4 VALID with READY handshake

Handshake Dependencies

- The VALID signal of one AXI component must be independent on the READY signal of the other component in the transaction. (prevent deadlock)
- The READY signal can wait for assertion of the VALID signal.

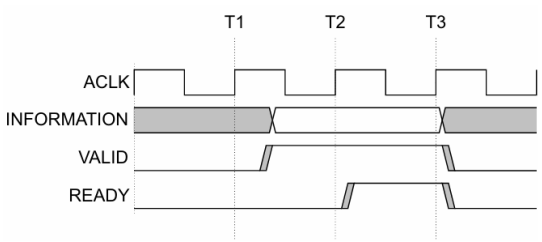


Figure A3-2 VALID before READY handshake

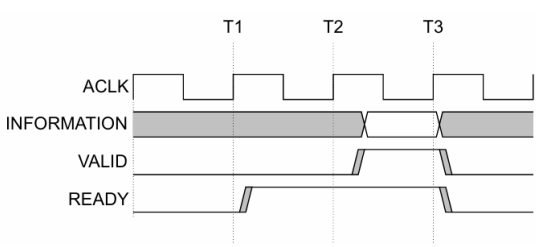


Figure A3-3 READY before VALID handshake

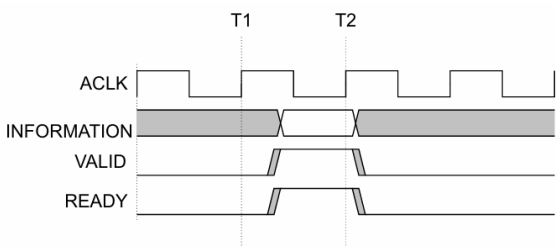
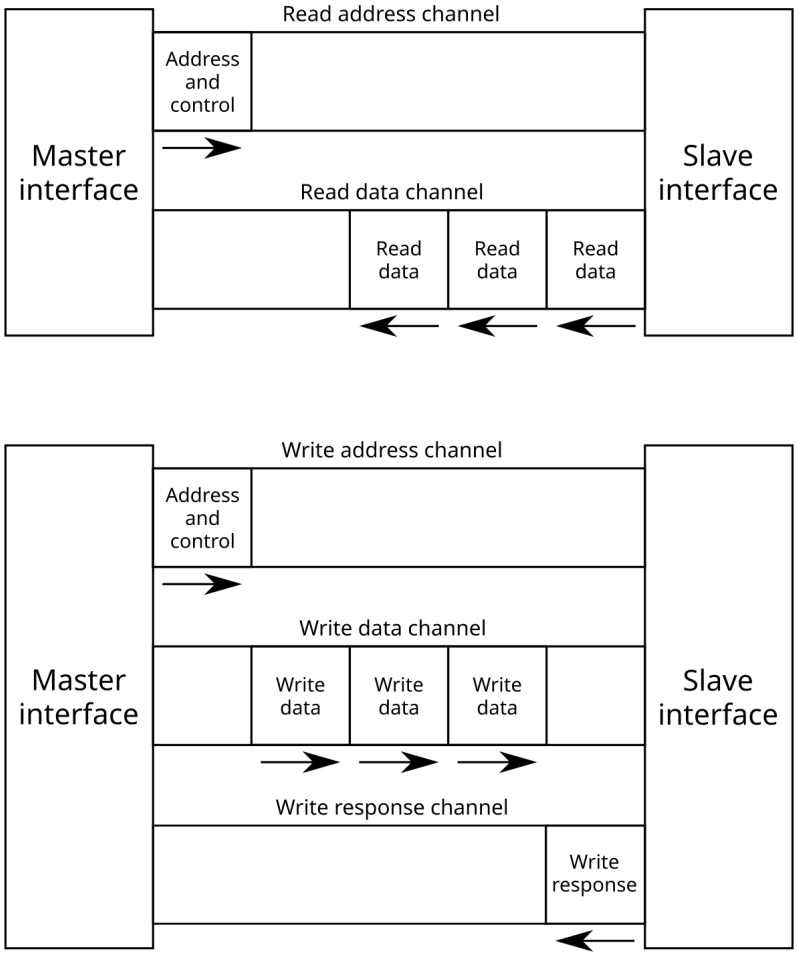
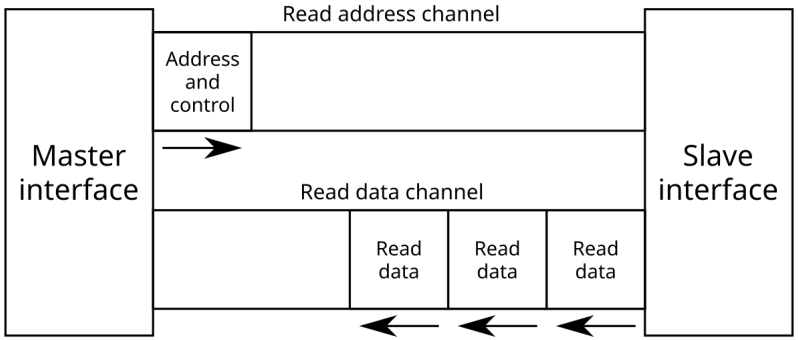


Figure A3-4 VALID with READY handshake

AXI Read/Write Channel



AXI Read/Write Channel



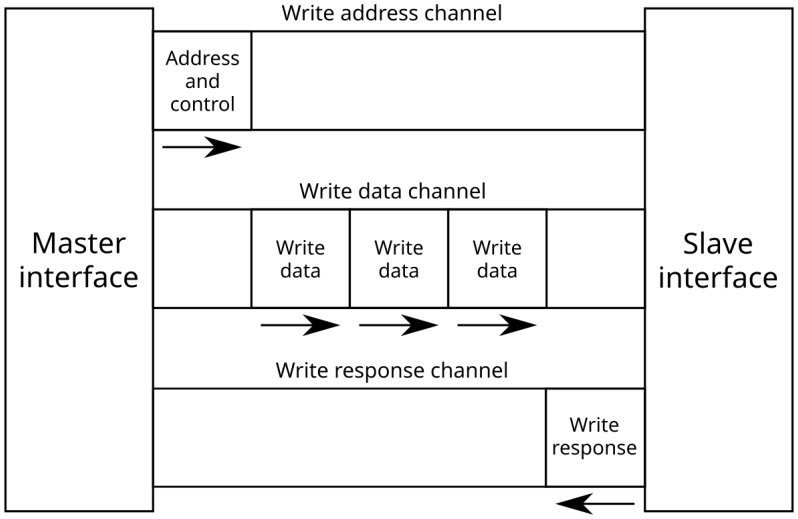


Table A3-1 Transaction channel handshake pairs

Transaction channel	Handshake pair
Write address channel	**AWVALID, AWREADY**
Write data channel	**WVALID, WREADY**
Write response channel	**BVALID, BREADY**
Read address channel	**ARVALID, ARREADY**
Read data channel	**RVALID, RREADY**

AXI Signal Description

- AXI Transaction：the complete set of required operations on the AXI bus form the AXI Transaction
- AXI Burst：any required payload data is transferred as an AXI Burst
- AXI Beats：a burst can comprise multiple data transfers, or AXI Beats

ARLEN	Master	Burst length. This signal indicates the exact number of transfers in a burst. This changes between AXI3 and AXI4. See *Burst length* on page A3-44.
ARSIZE	Master	Burst size. This signal indicates the size of each transfer in the burst. See *Burst size* on page A3-45.
ARBURST	Master	Burst type. The burst type and the size information determine how the address for each transfer within the burst is calculated. See *Burst type* on page A3-45.

Table A3-3 Burst type encoding

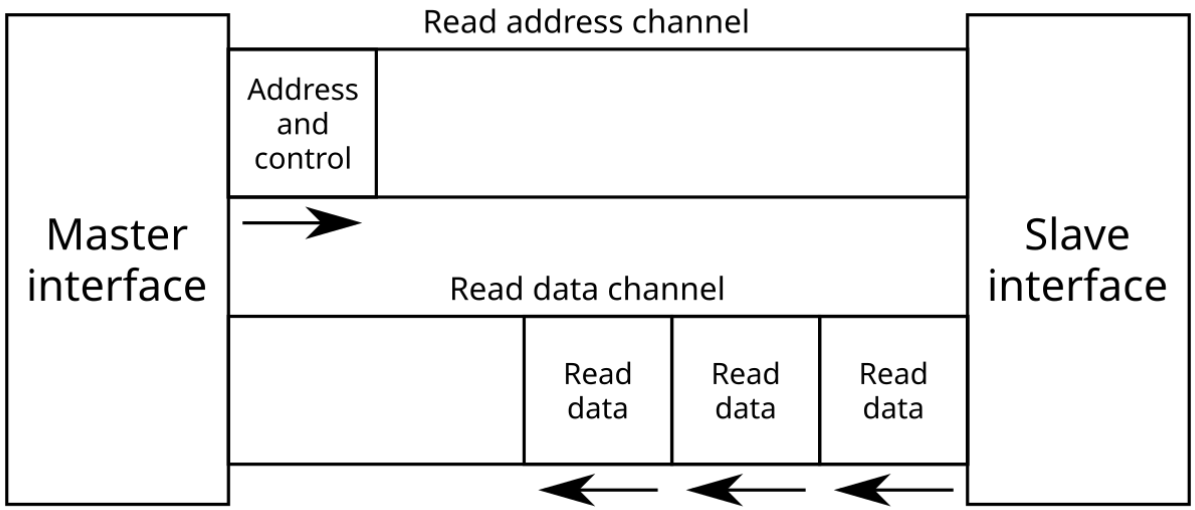
AxBURST[1:0]	Burst type
0b00	FIXED
0b01	INCR
0b10	WRAP
0b11	Reserved

Table A3-2 Burst size encoding

AxSIZE[2:0]	Bytes in transfer
0b000	1
0b001	2
0b010	4
0b011	8
0b100	16
0b101	32
0b110	64
0b111	128

AXI 4 Basic Transaction

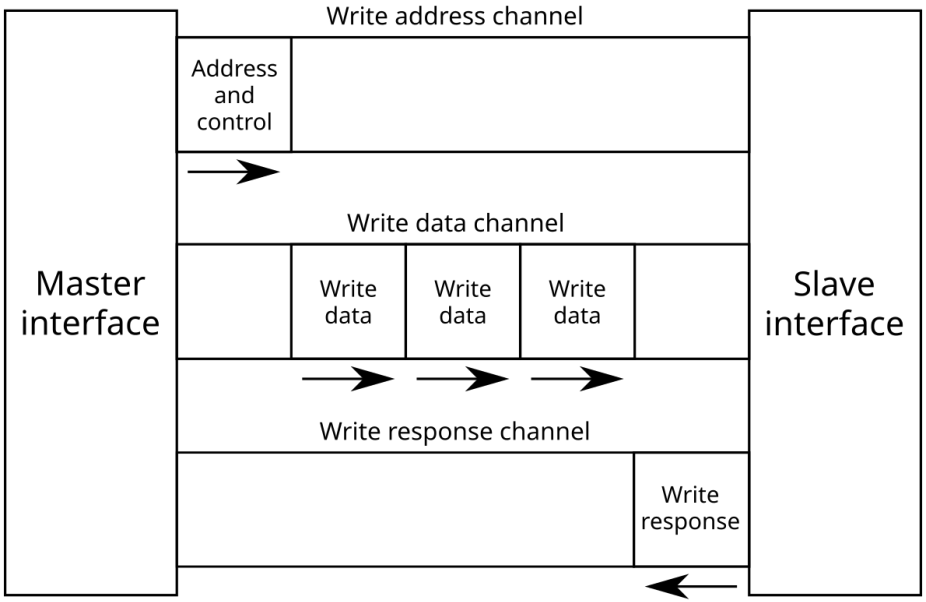
- Read Transaction
  - Read Address Channel
  - Read Data Channel



- Each channel have valid-ready pair for handshaking process

AXI 4 Basic Transaction

- Write Transaction
  - Write Address Channel
  - Write Data Channel
  - Write Response Channel



- Each channel have **valid-ready pair** for handshaking process

Global Signals

Signal	Source	Description
clk	Clock source	Global clock signal. All signals are sampled on the rising edge of the global clock.
rst_n	Reset source	Global reset signal. This signal is active LOW.

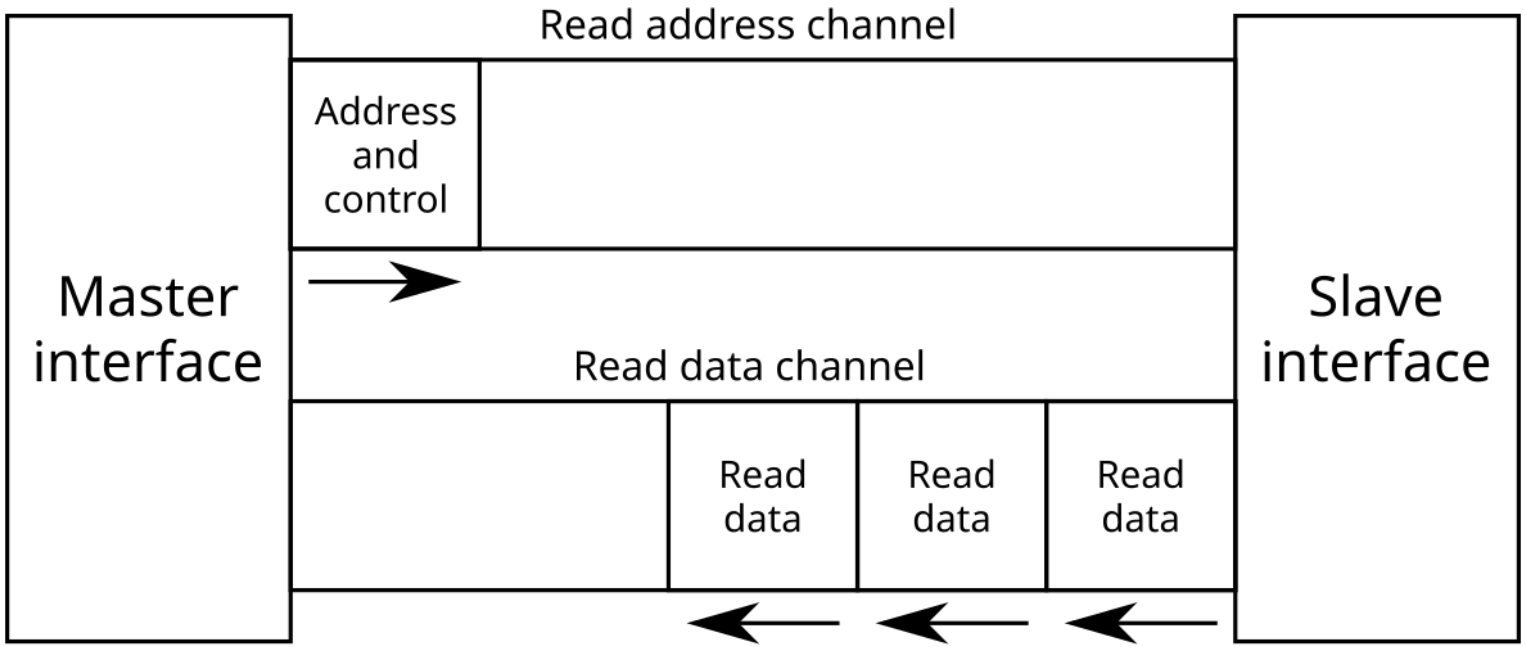
Read Address Channel

Signal	Source	Description
ARID[3:0]	Master	Read address ID. This signal is the identification tag for the read address group of signals.
ARADDR[31:0]	Master	Read address. The read address gives the address of the first transfer in a read burst transaction.
ARLEN[7:0]	Master	Burst length. The burst length gives the exact number of transfers in a burst. This information determines the number of data transfers associated with the address.
ARSIZE[2:0]	Master	Burst size. This signal indicates the size of each transfer in the burst.
ARBURST[1:0]	Master	Burst type. The burst type, coupled with the size information, details how the address for each transfer within the burst is calculated.
ARVALID	Master	Read address valid. This signal indicates, when HIGH, that the read address and control information is valid and will remain stable until the address acknowledge signal, **ARREADY**, is high. 1 = address and control information valid 0 = address and control information not valid.
ARREADY	Slave	Read address ready. This signal indicates that the slave is ready to accept an address and associated control signals: 1 = slave ready 0 = slave not ready.

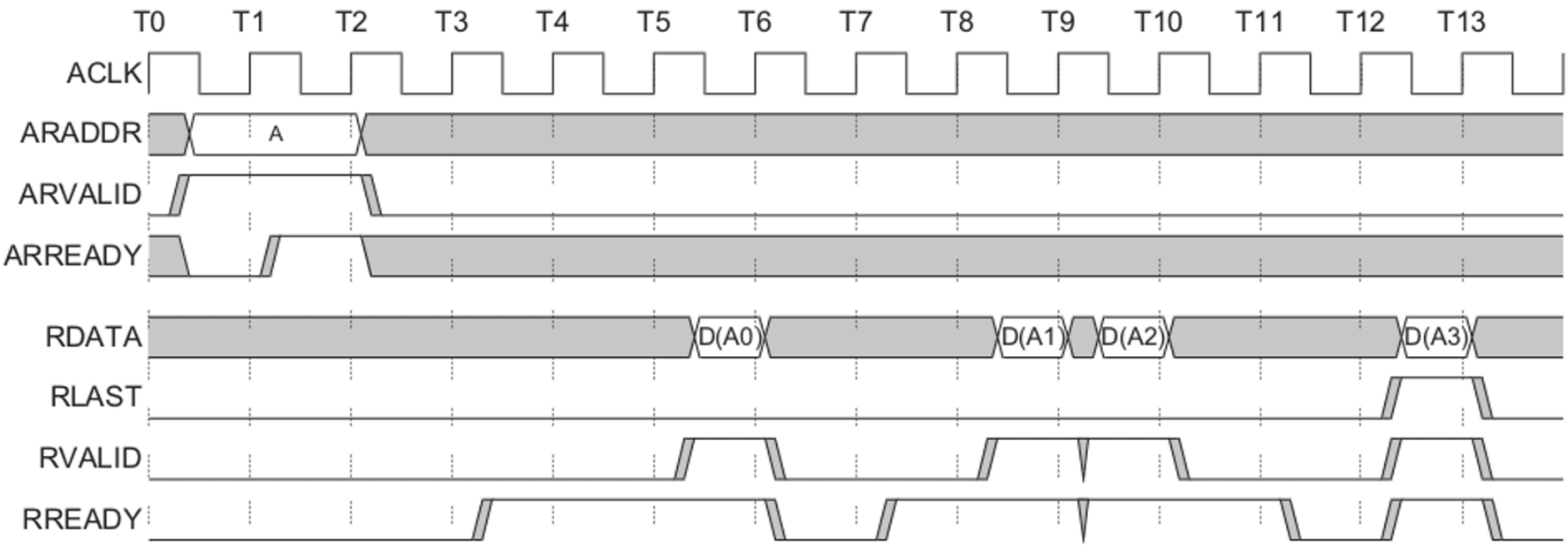
Read Data Channel

Signal	Source	Description
RID[3:0]	Slave	Read ID tag. This signal is the ID tag of the read data group of signals. The RID value is generated by the slave and must match the ARID value of the read transaction to which it is responding.
RDATA	Slave	Read data.
RRESP[1:0]	Slave	Read response. This signal indicates the status of the read transfer. The allowable responses are OKAY, EXOKAY, SLVERR, and DECERR.
RLAST	Slave	Read last. This signal indicates the last transfer in a read burst.
RVALID	Slave	Read valid. This signal indicates that the required read data is available and the read transfer can complete: 1 = read data available 0 = read data not available.
RREADY	Master	Read ready. This signal indicates that the master can accept the read data and response information: 1= master ready 0 = master not ready.

Read Transaction



Waveform of Read Transaction



Write Address Channel

Signal	Source	Description
AWID[3:0]	Master	Write address ID. This signal is the identification tag for the write address group of signals.
AWADDR[31:0]	Master	Write address. The write address bus gives the address of the first transfer in a write burst transaction. The associated control signals are used to determine the addresses of the remaining transfers in the burst transaction.
AWLEN[7:0]	Master	Burst length. The burst length gives the exact number of transfers in a burst. This information determines the number of data transfers associated with the address.
AWSIZE[2:0]	Master	Burst size. This signal indicates the size of each transfer in the burst.
AWBURST[1:0]	Master	Burst type. The burst type, coupled with the size information, details how the address for each transfer within the burst is calculated.
AWVALID	Master	Write address valid. This signal indicates that valid write address and control information are available: 1 = address and control information available 0 = address and control information not available. The address and control information remain stable until the address acknowledge signal, AWREADY, goes HIGH.
AWREADY	Slave	Write address ready. This signal indicates that the slave is ready to accept an address and associated control signals: 1 = slave ready 0 = slave not ready.

Write Data Channel

Signal	Source	Description
WDATA	Master	Write data. The write data bus can be 8, 16, 32, 64, 128, 256, 512, or 1024 bits wide.
WLAST	Master	Write last. This signal indicates the last transfer in a write burst.
WVALID	Master	Write valid. This signal indicates that valid write data and strobes are available: 1 = write data and strobes available 0 = write data and strobes not available.
WREADY	Slave	Write ready. This signal indicates that the slave can accept the write data: 1 = slave ready 0 = slave not ready.

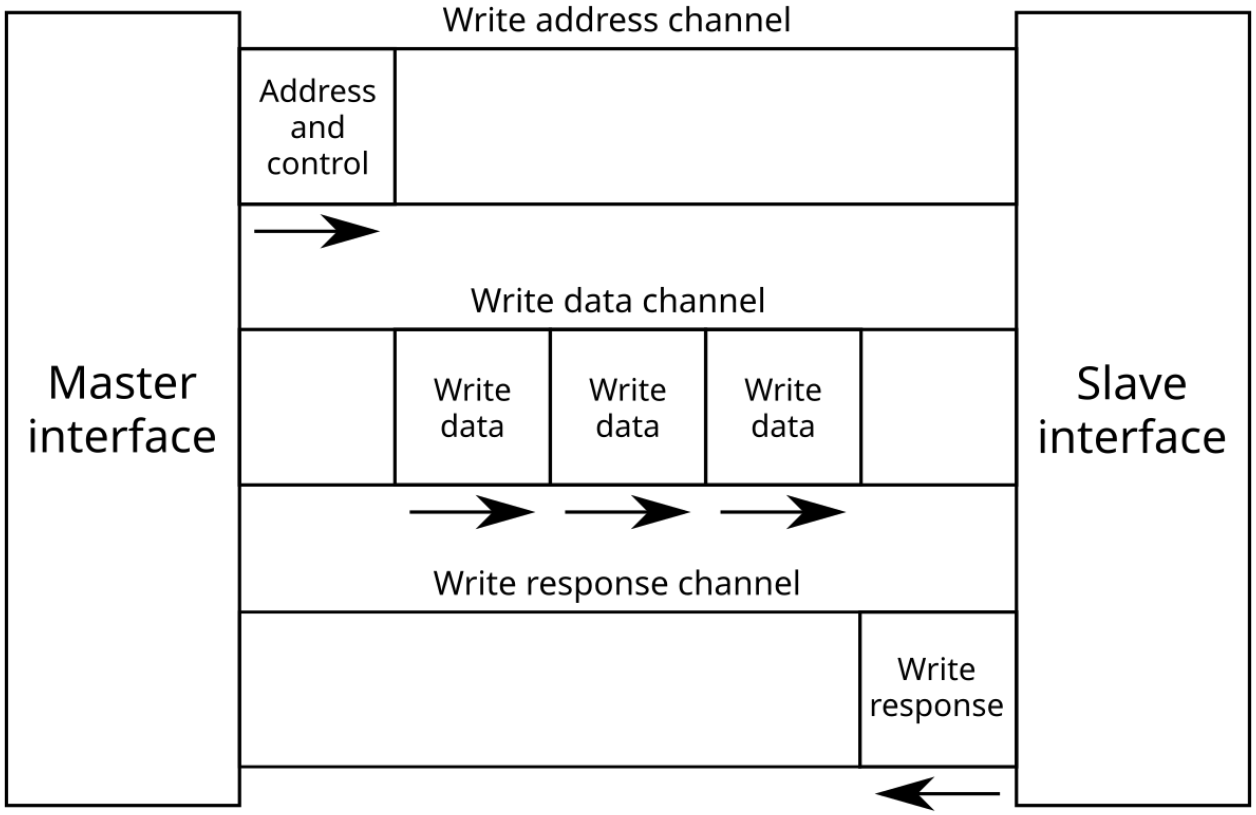
Write Response Channel

Signal	Source	Description
BID[3:0]	Slave	Response ID. The identification tag of the write response. The BID value must match the AWID value of the write transaction to which the slave is responding.
BRESP[1:0]	Slave	Write response. This signal indicates the status of the write transaction. The allowable responses are OKAY, EXOKAY, SLVERR, and DECERR.
BVALID	Slave	Write response valid. This signal indicates that a valid write response is available: 1 = write response available. 0 = write response not available.
BREADY	Master	Response ready. This signal indicates that the master can accept the response information. 1 = master ready. 0 = master not ready.

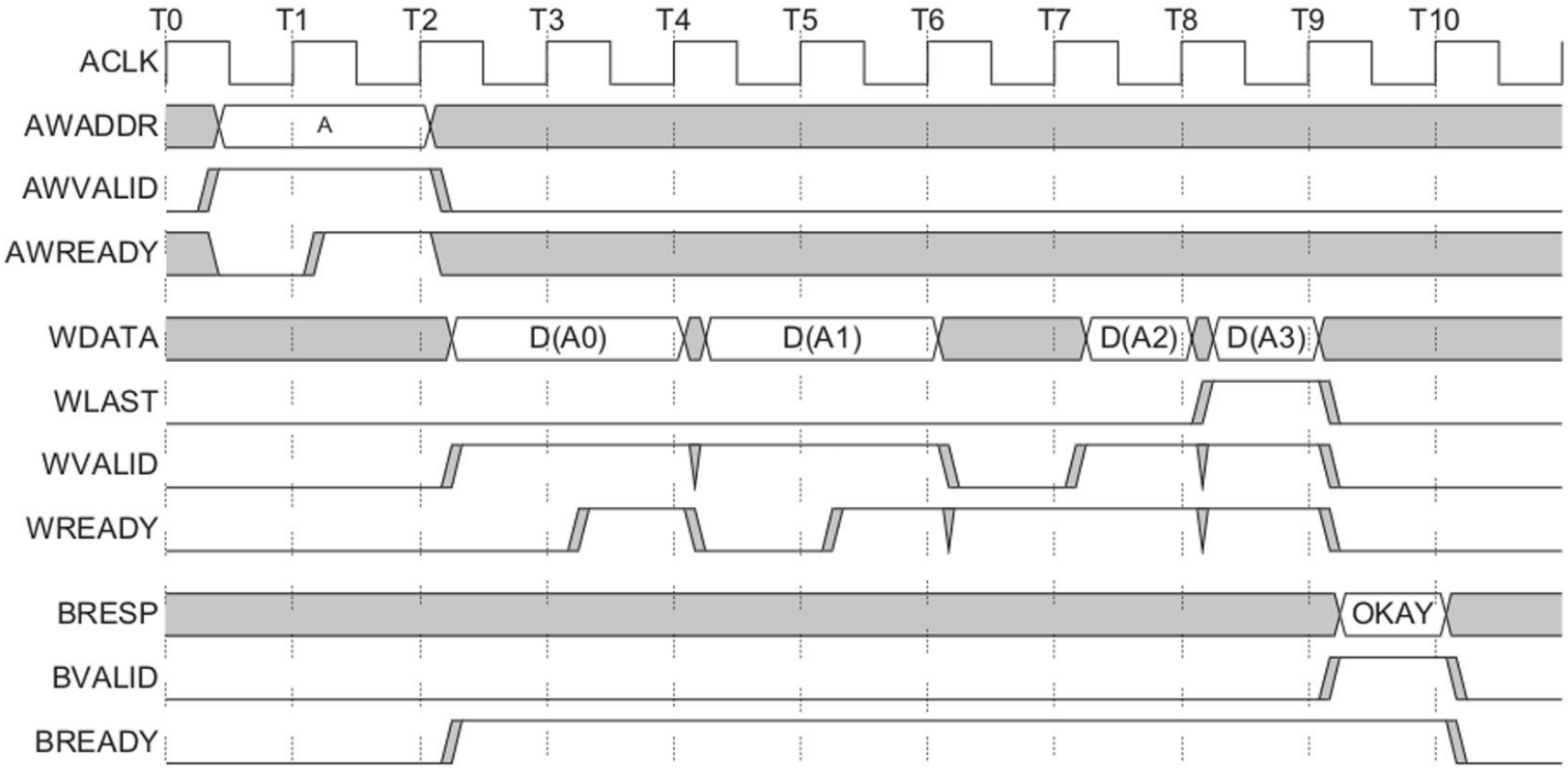
Table A3-4 RRESP and BRESP encoding

RRESP[1:0] BRESP[1:0]	Response
`0b00`	OKAY
`0b01`	EXOKAY
`0b10`	SLVERR
`0b11`	DECERR

Write Transaction



Waveform of Write Transaction



Reference

- AMBA AXI and ACE Protocol Specification AXI3, AXI4, and AXI4-Lite ACE and ACE-Lite
- Design of Multi Master Single Slave AXI4